F. Task(s)

Do a research project with some creativity in solving problem using application of computer vision in one of these domains (but not limited to):

1- Transportation:

Develop a computer vision model that can detect and classify objects in real-time, such as pedestrians, other vehicles, and road signs, to aid in the development of autonomous driving technology. You will build your own detection framework and try it on a standard benchmark dataset. Example research problems include how to balance between speed and accuracy, and how object scale, density or lightning condition affect the detection result.

2- Healthcare

Create a model that can analyze medical images, such as X-rays or MRI scans, to aid in the diagnosis or detection of diseases or medical conditions.

3- Robotics

Build a computer vision model that enables a robot to recognize and interact with objects in its environment, allowing it to perform tasks such as object manipulation or navigation.

4- Agriculture

Develop a model that can analyze images of crops or fields to monitor growth, detect diseases, or identify pests, allowing for more efficient and effective agricultural practices.

5- Sports

Create a model that can track players' movements and actions during games. They can extract data on player positions, speed, trajectories, and interactions, enabling detailed performance analysis and strategic insights for coaches and athletes.

6- Manufacturing

Create a computer vision model that can monitor production lines in real-time. They can analyze visual data to detect anomalies, identify bottlenecks, or track the flow of materials or products. This allows manufacturers to optimize production processes, improve throughput, and minimize downtime.

The phases of the project shall include:

1- Do literature review about the application and different algorithms that have been applied there.
2- Find benchmark dataset.
3- Building and train your model (you may consider the existing version of any of the papers you have reviewed in phase 1 but you need to do some comparison by enhancing the model or finetuned the parameters).
4- Test and evaluate your model.
5- Write your report.

Your report shall be in a format of conference paper (Computer Vision and pattern recognition CVPR) and shall have these sections:

1- Abstract: To summarize your project include problem, solution and result.
2- Introduction: To introduce the topic to the reader combined with the related work in the domain and problem you will solve.
3- Methodology: To present the method that you will use including description of the dataset.
4- Result and discussion: To present the result analysis and discuss it including limitation and future work.
5- Conclusion: To conclude what you have done.
6- References: List of references (at least 8).

Turnitin in similarity report must be below 20 % (No need to include the report in the submission., However you need to check the similarity using the given enrolment code).